EECE 306 Lab 6 notebook. Electric Potential, the Gradient, and Path Independence

Team 3: Amelia Harris, Carissa McWilliams, Mitch Nazareth. TODO replace with your team number and member names.

Fill in every TODO, run `publish('Lab06_notebook.m')` from the toolbox root, print the HTML to PDF, three to five pages.

Contents

- Setup
- Potential against the closed form
- E from the gradient, and the error map
- Choosing the step size, the V shaped curve
- The field is conservative, with a control
- Potential difference two ways
- Seeing the potential
- Interpretation
- Problems encountered
- Full test suite

Setup

```
clear; close all;
e0 = em.const.eps0();
Q  = 2e-9;
s  = em.src.pointCharge(Q, [0 0 0]);
FE = @(r) em.field.E(s, r);
FV = @(r) em.field.V(s, r);
```

Potential against the closed form

```
radii = (1:10)';
Vnum = em.field.V(s, [radii zeros(10,2)]);
Vref = Q ./ (4.*pi.*e0.*radii);
disp('   r (m)    relative error')
disp([radii abs(Vnum - Vref)./Vref])
em.test.assertClose(Vnum, Vref, 1e6, 'Potential against the closed form')
```

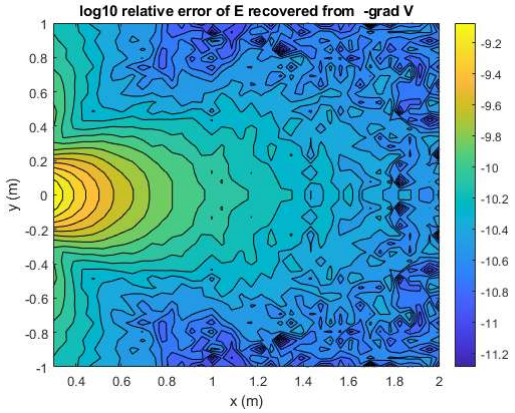
```
   r (m)    relative error
    1.0000         0
    2.0000         0
    3.0000         0
    4.0000         0
    5.0000    0.0000
    6.0000         0
    7.0000         0
    8.0000         0
    9.0000         0
   10.0000    0.0000
```

E from the gradient, and the error map

E computed directly and E recovered as minus the gradient of V must agree away from the source.

```
[X, Y] = meshgrid(linspace(0.3, 2, 40), linspace(-1, 1, 40));
P = [X(:) Y(:) zeros(numel(X),1)];
Ed = em.field.E(s, P);
Eg = -em.op.grad(FV, P);
em.test.assertClose(Ed, Eg, 1e-6, ['Electric field from gradient and' ...
    'direct calculation'])
errmap = reshape(em.vec.mag(Ed - Eg) ./ em.vec.mag(Ed), size(X));
figure; contourf(X, Y, log10(errmap), 20); colorbar
xlabel('x (m)'); ylabel('y (m)')
title('log10 relative error of E recovered from  -grad V')
% TODO one or two sentences on where the error is largest and why.

% Error is largest near the edge of the region where the gradient
% is calculated because the gradient is approximated using finite
% difference. This creates relative error that gets bigger when
% the electric field is smaller.
```



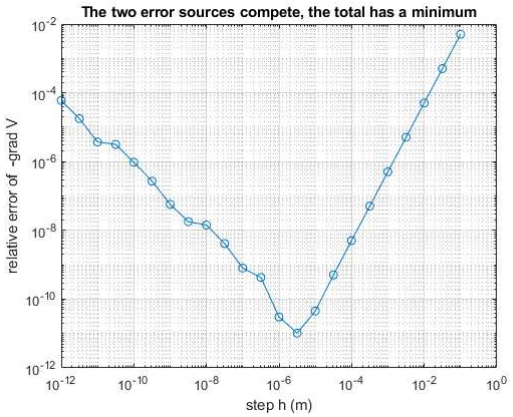
Choosing the step size, the V shaped curve

Truncation error falls as h squared, round off grows as 1 over h. The total has a minimum. Sweep h and find it for this machine.

```
p0 = [1 0.5 0];
Eexact = em.field.E(s, p0);
hs = logspace(-12, -1, 23)';
errh = zeros(size(hs));
```

```
for k = 1:numel(hs)
    Eh = -em.op.grad(FV, p0, hs(k));
    errh(k) = em.vec.mag(Eh - Eexact) / em.vec.mag(Eexact);
end
figure; loglog(hs, errh, 'o-'); grid on
xlabel('step h (m)'); ylabel('relative error of  -grad V')
title('The two error sources compete, the total has a minimum')
[emin, imin] = min(errh);
fprintf('best h on this machine about %.1e with error %.1e\n', hs(imin), emin);
fprintf('the API default h = 1e-5 sits on the flat bottom of this curve\n');
```

```
best h on this machine about 3.2e-06 with error 1.0e-11
the API default h = 1e-5 sits on the flat bottom of this curve
```



The field is conservative, with a control

```
loops = { @(t)[2.*cos(t) 2.*sin(t) zeros(size(t))], ...
          @(t)[1+0.5.*cos(t) 0.5*sin(3*t) 0.3.*sin(t)], ...
          @(t)[0.5.*cos(t) 1+0.8.*sin(t) 0.2.*cos(2.*t)] };
for k = 1:3
    C = em.field.circulation(FE, loops{k}, [0 2*pi], 400);
    fprintf('circulation of E on path %d = %.3e\n', k, C);
end
Fnc = @(r) [-r(:,2) r(:,1) zeros(size(r,1),1)];
Cnc = em.field.circulation(Fnc, loops{1}, [0 2*pi], 400);
fprintf('circulation of the control field = %.4f  (expect 8 pi, clearly nonzero)\n', Cnc);
em.test.assertClose(Cnc, 8*pi, 1e-6, 'Circulation test')
% The control matters. A test that only ever confirms zero cannot tell
% correct code from code that always returns zero.
```

```
circulation of E on path 1 = 8.299e-16
circulation of E on path 2 = 4.611e-10
circulation of E on path 3 = -4.675e-11
circulation of the control field = 25.1327  (expect 8 pi, clearly nonzero)
```

Potential difference two ways

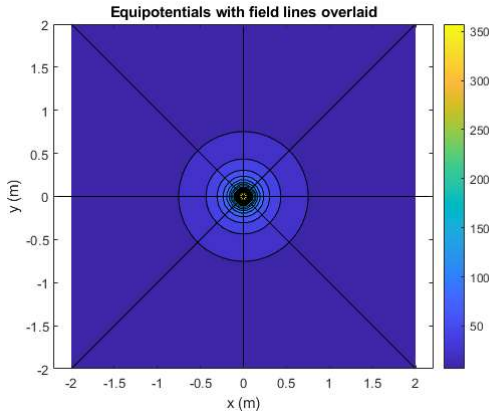
```
pa = [1 0 0]; pb = [0.3 0.4 0];
dV = em.field.V(s, pb) - em.field.V(s, pa);
seg = @(t) pa + t.*(pb - pa);
work = -em.field.circulation(FE, seg, [0 1], 400);
fprintf('V(b) - V(a) from em.field.V        = %.6e V\n', dV);
fprintf('minus line integral of E, a to b   = %.6e V\n', work);
```

```
V(b) - V(a) from em.field.V        = 1.797510e+01 V
minus line integral of E, a to b   = 1.797513e+01 V
```

Seeing the potential

```
figure;
em.viz.contourV(@(r) em.field.V(s, r), [-2 2], [-2 2], 60, struct('title', 'Equipotentials of a point charge'));
hold on
em.viz.streamlines(FE, [-2 2], [-2 2], [cosd(0:45:315)' sind(0:45:315)' zeros(8,1)].*0.2, struct());
title('Equipotentials with field lines overlaid')
```

```
size(Vval) = [3600 1]
min(Vval) = 6.35516
max(Vval) = 374.954
NaNs = 0
Infs = 0
```



Interpretation

TODO three to six sentences. State what the step size study implies for every later use of em.op.grad, and why driving h smaller is not safer.

```
% The step size study implies that there is a size of h small enough to
% ensure the best precision without worsening the error caused by
% round-off. This error is caused by a loss in numerical precision
% when calculating the difference between two very close
% potential values. Driving h to be smaller is not always
% safer because eventually the two numerical values become
% close enough that the difference between them loses precision.
```

Problems encountered

TODO honest account, or NONE.

```
% The midpoint quadrature and numerical derivative caused the relative
% error to be larger than the expected relative error. Switching to Gauss's
% equations resolved the issue. There were also issues with laying the
% field lines over the quiver2 plot, and it was diffult to determine how
% em.field.V would test if a source has an extent that grows with N. The
% original notebook did not use element-wise multiplication in the function
% handles.
```

Full test suite

```
runTests
```

```
EECE 306 toolbox test suite
------------------------------------------------------------------
24.2.0.3212159 (R2024b) Update 9
eps0 = 8.8541878128e-12 F/m
mu0  = 1.2566370614e-06 H/m
c0   = 2.9979245808e+08 m/s
eta0 = 376.730314 ohm
Derived constant checks passed.
mag([3 4 0])  = 5   (expect 5 exactly)
unit([3 4 0]) =
    0.6000    0.8000         0

unit([0 0 0]) (expect all NaN) =
   NaN   NaN   NaN

angle perpendicular = 1.5707963267949  (expect pi/2)
angle antiparallel  = 3.14159265358979, real = 1  (expect pi, 1)
fromTo([1 1 1],[2 3 4]) =
     1     2     3

mag    input 100x3 output [100 1]
unit   input 100x3 output [100 3]
angle  input 100x3 output [100 1]
fromTo input 100x3 output [100 3]
  PASS  test_lab01            0.51 s
spherical   roundtrip max error = 6.357e-15
cylindrical roundtrip max error = 2.979e-15
c2sph([0 0 1])  =
     1     0     0

c2sph([0 0 -1]) =
    1.0000    3.1416         0

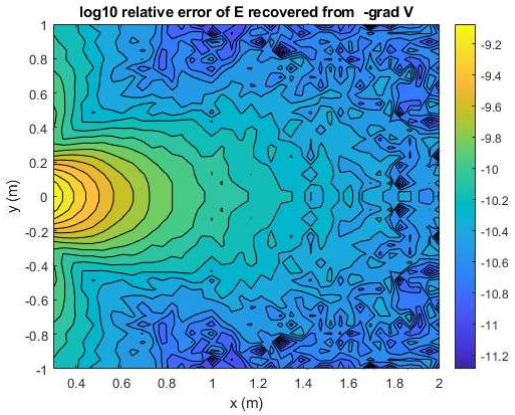
phi in the four quadrants = 0.7854 2.3562 3.9270 5.4978 rad
all inside [0, 2*pi) = 1
vector roundtrip (spherical pair) max error = 1.088e-15
magnitude change under conversion max       = 8.882e-16
vector roundtrip (cylindrical pair) max error = 6.280e-16
at (0, 5, 0), (Ar, Atheta, Aphi) =
    3.0000    0.0000   -2.0000

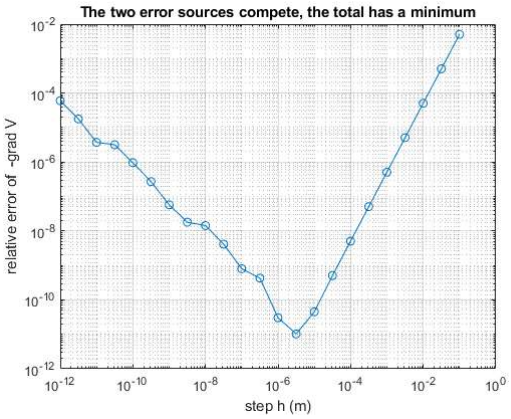
at (4, 0, 0), (Ar, Atheta, Aphi) =
    2.0000    0.0000    3.0000

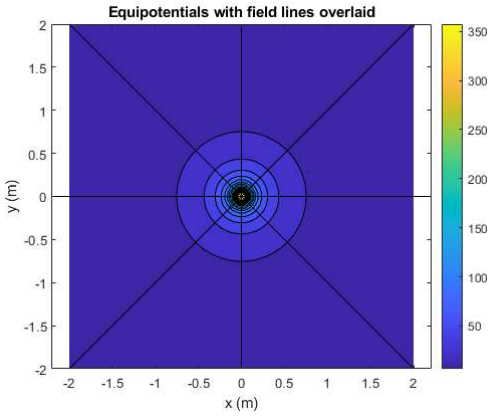
  PASS  test_lab02            0.28 s
   r (m)    relative error
    1.0000         0
    2.0000         0
    3.0000         0
    4.0000         0
    5.0000         0
    6.0000         0
    7.0000         0
    8.0000         0
    9.0000    0.0000
   10.0000         0
```

```
Minimum charge of positive charges = 7.463e-01 (should be greater than 0)
Maximum charge of negative charges = -7.463e-01 (should be less than 0)
max |Ex| on bisector      = 0.000e+00 V/m
max |E| on the same plane = 5.061e+02 V/m
element count after first merge = 2 (expect 2)
Mismatch test passed: em.src.merge(q1, crnt_struct) successfully raised an error.
|E(r)| / |E(2r)| = 8.0003   (expect near 8 for a dipole)
Singularity test passed: em.field.E(q, [0 0 0]) successfully raised an error.
em.field.E on 5000 points took 0.008 s  (requirement, under 2 s)
Observation points: N = 10, Source Charges: M = 1, Product = 10, Time = 0.000 s
Observation points: N = 100, Source Charges: M = 1, Product = 100, Time = 0.000 s
Observation points: N = 1000, Source Charges: M = 1, Product = 1000, Time = 0.000 s
Observation points: N = 10000, Source Charges: M = 1, Product = 10000, Time = 0.002 s
Observation points: N = 100000, Source Charges: M = 1, Product = 100000, Time = 0.018 s
Observation points: N = 1000000, Source Charges: M = 1, Product = 1000000, Time = 0.132 s
Observation points: N = 10000000, Source Charges: M = 1, Product = 10000000, Time = 1.188 s
Compution time surpasses time threshold at N=100000000 points: took 18.753 seconds
  PASS  test_lab03           25.84 s
circle length  6.283185   expect 6.283185
sphere area    12.567806   expect 12.566371
ball volume    4.187181   expect 4.188790
square area    1.000000   expect 1.000000
midpoint  observed order p = 2.00
trapz     observed order p = 2.09
simpson   observed order p = 4.00
gauss N =  2   error = 3.855e-04
gauss N =  4   error = 9.330e-10
gauss N =  8   error = 6.661e-16
gauss N = 16   error = 2.887e-15
ratio first exceeds 0.99 at L/d about 19.3
ratio first exceeds 0.99 at a/h about 59.8
net charge of rho(z) = z on [-1, 1] = -0.000 C  (expect 0)
  PASS  test_lab04            3.59 s
flux through sphere   = 3.000000e-09 C
enclosed charge       = 3.000000e-09 C
relative error        = 4.175e-08
flux with charge outside = 0.000 C  (consistent with zero)
sphere               3.000000e-09 C
cube                 3.000204e-09 C
hemisphere and cap  3.000001e-09 C
charge at x = 0.00   flux = 3.000000e-09 C
charge at x = 0.15   flux = 3.000000e-09 C
charge at x = 0.30   flux = 3.000000e-09 C
Nu = 10   div D inside = 1.0163e-06   rho_v = 1.0000e-06   rel err = 1.63e-02
Nu = 20   div D inside = 1.6814e-06   rho_v = 1.0000e-06   rel err = 6.81e-01
Nu = 40   div D inside = 7.4061e-07   rho_v = 1.0000e-06   rel err = 2.59e-01
div D in charge free space = 0.000  (consistent with zero)
flux with swapped parameter order = -3.000000e-09 C  (sign flipped)
  PASS  test_lab05            1.37 s
   r (m)    relative error
    1.0000         0
    2.0000         0
    3.0000         0
    4.0000         0
    5.0000    0.0000
    6.0000         0
    7.0000         0
    8.0000         0
    9.0000         0
   10.0000    0.0000

best h on this machine about 3.2e-06 with error 1.0e-11
the API default h = 1e-5 sits on the flat bottom of this curve
circulation of E on path 1 = 8.299e-16
circulation of E on path 2 = 4.611e-10
circulation of E on path 3 = -4.675e-11
circulation of the control field = 25.1327  (expect 8 pi, clearly nonzero)
V(b) - V(a) from em.field.V        = 1.797510e+01 V
minus line integral of E, a to b   = 1.797513e+01 V
size(Vval) = [3600 1]
min(Vval) = 6.35516
max(Vval) = 374.954
NaNs = 0
Infs = 0
  PASS  test_lab06            0.97 s
------------------------------------------------------------------
  6 passed, 0 failed, 6 total
  All tests passing.
```







Published with MATLAB® R2024b